

Neural Surrogates for Path-Dependent Volatility: Fast Online Assimilation of Streaming Option Quotes with Guardrails

When constructing precise models for option prices, one must consider time-varying volatility and history-dependent behavior. There was an interesting idea presented in Guyon & Lekeufack [1], who developed the Path-Dependent Volatility (PDV) framework wherein volatility depended on prior returns on assets. Although the Monte Carlo method can price options and numerically calibrate this model effectively, it was computationally intensive. Recent literature indicated that neural networks may transfer the burden of volatility calibration from prediction time to training time [2]. My main intention was to investigate whether the neural network might be used to approximate the calibration procedure in less time with protection against incorrect predictions. In addition, I used an inverse neural surrogate that received an option-price surface and current PDV state as an input to predict five calibration parameters, with numerical calibration remaining as a backup approach in case the neural surrogate failed any statistical or numerical guardrail test.

Quotes for SPX and SPX Weekly options were gathered daily from ThetaData at approximately 3:55 PM ET on twenty trading days between May 5 and June 2, 2021. The quotes were grouped according to seven maturities close to 7, 14, 21, 30, 45, 60, and 90 days and then interpolated along eleven log-moneyness points. Thus, each surface consisted of seventy-seven normalized option prices. Next, I used a restricted four-factor Markovian approximation of the non-Markovian PDV model described above, with five parameters: β_0 , β_1 , β_2 , θ_1 , and θ_2 along with four PDV states calculated based on the recent SPX return history. Pricing and calibration procedures were performed using the Monte Carlo method [3]. Each historical configuration was

calibrated using 4,000 paths and independently priced using 20,000 paths across three different random seeds. Four configurations failed to generate positive volatility regardless of convergence; hence, there were sixteen stable empirical anchors. A total of ten thousand bounded perturbations were created around the sixteen anchors while maintaining their associated states; invalid solutions were excluded, yielding 9,667 synthetic examples.

These scenarios were made up of seventy-seven option prices and four PDV states, resulting in eighty-one input features with five predicted parameters. The final neural architecture took the form of 81 -> 256 -> 256 -> 128 -> 5 using GELU activation functions with standardized inputs and outputs. Testing this model on an independent 959-example in-domain test set spanning all sixteen state neighborhoods yielded R^2 scores ranging from 0.9689 to 0.9943 across the five parameters. Predictions exceeding parameter boundaries were projected into a valid range prior to pricing. Out of 959 scenarios, 955 were able to pass full PDV simulation with mean surface RMSE of 1.41 basis points of forward and median RMSE of 1.05 basis points. In another experiment, sixty-four balanced test scenarios were tested under three additional random seeds; all of them were successful in completing 192 comparisons with mean surface RMSE of 1.32 basis points. The median latency of ANN forward passes was 0.109 milliseconds, with 99th-percentile latency around 0.120 milliseconds.

However, the major issue occurred during out-of-domain testing wherein recovering correct parameters deteriorated significantly. To address the problem, I designed an out-of-distribution detection technique relying on Mahalanobis distance in the eighty-one-dimensional space of inputs. Ledoit-Wolf shrinkage was used to stabilize the estimated covariance matrix before computing Mahalanobis distances. Using a 99th-percentile threshold based on the held-out in-domain examples yielded approximately 1.01% false

positives, along with detecting 100% out-of-domain data drawn from May 27, May 28, June 1, and June 2. Therefore, the last guarded architecture employed ANN only on in-domain inputs, projected parameters to a feasible interval, and discarded numerically incorrect predictions to fall back on numerical calibration procedure. Overall, the results showed accurate parameter recovery, efficient reproduction of option surfaces with 1-2 basis points of error on average within the domain of the model, and necessity of explicit guardrails outside the domain.

As mentioned above, my initial objective included combining a computationally expensive reference PDV model with fast ANN and evaluating the trade-off between accuracy, stability, and efficiency. The work succeeded in accomplishing these primary objectives despite changes made to methodology throughout the process. Namely, it was transformed into training an inverse neural surrogate based on synthetic surfaces derived from actual market data. Furthermore, I obtained sub-millisecond neural inference, basis point-level re-pricing accuracy, and an insight into guardrails' significance beyond the domain boundary. Further investigation might include adapting this framework for intraday data analysis, studying the distribution of pricing errors across strikes, maturities, and market conditions, and benchmarking ANN against optimized numerical calibration techniques.

Personally, participating in UROP allowed me to understand how real-world research progresses with new discoveries contradicting initial assumptions. Besides learning about building data pipelines for financial instruments and running large-scale Monte Carlo simulations on the MSI supercomputing cluster, I gained valuable experience with ANN training and validation. Above everything else, failure cases turned into crucial insights, which were directly utilized in guardrail development.

References

- [1] J. Guyon and J. Lekeufack, “Volatility is (mostly) path-dependent,” *Quantitative Finance*, vol. 23, no. 9, pp. 1221-1258, 2023.
- [2] B. Horvath, A. Muguruza, and M. Tomas, “Deep Learning Volatility: A deep neural network perspective on pricing and calibration in (rough) volatility models,” *Quantitative Finance*, vol. 21, no. 1, pp. 11-27, 2021.
- [3] P. Glasserman, *Monte Carlo Methods in Financial Engineering*. New York, NY, USA: Springer, 2003.

Appendix A: Selected Experimental Results

Table A1. Summary of inverse-surrogate and guardrail performance.

Metric	Result
Valid synthetic scenarios	9,667 / 10,000
β_0 test R^2	0.9934
β_1 test R^2	0.9883
β_2 test R^2	0.9943
θ_1 test R^2	0.9689
θ_2 test R^2	0.9933
Mean surface repricing RMSE	1.41 bp of forward
Fresh-seed mean repricing RMSE	1.32 bp of forward
Median ANN inference latency	0.109 ms
81-D chronological OOD detection	100%
In-domain p99 guardrail flag rate	1.01%

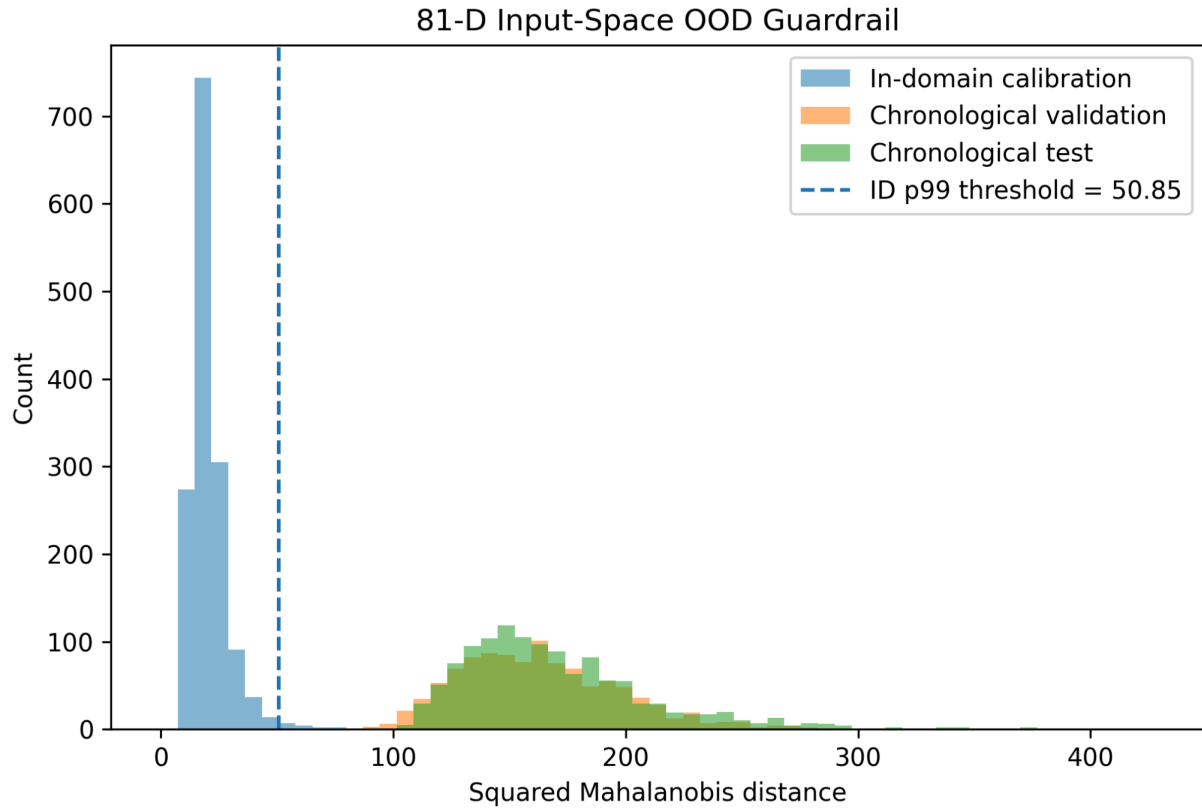


Figure A1. Mahalanobis-distance distributions for held-out in-domain observations and later chronological observations. The p99 in-domain threshold of 50.85 flagged all May 27-June 2 chronological observations as out-of-distribution while flagging approximately 1% of held-out in-domain observations.

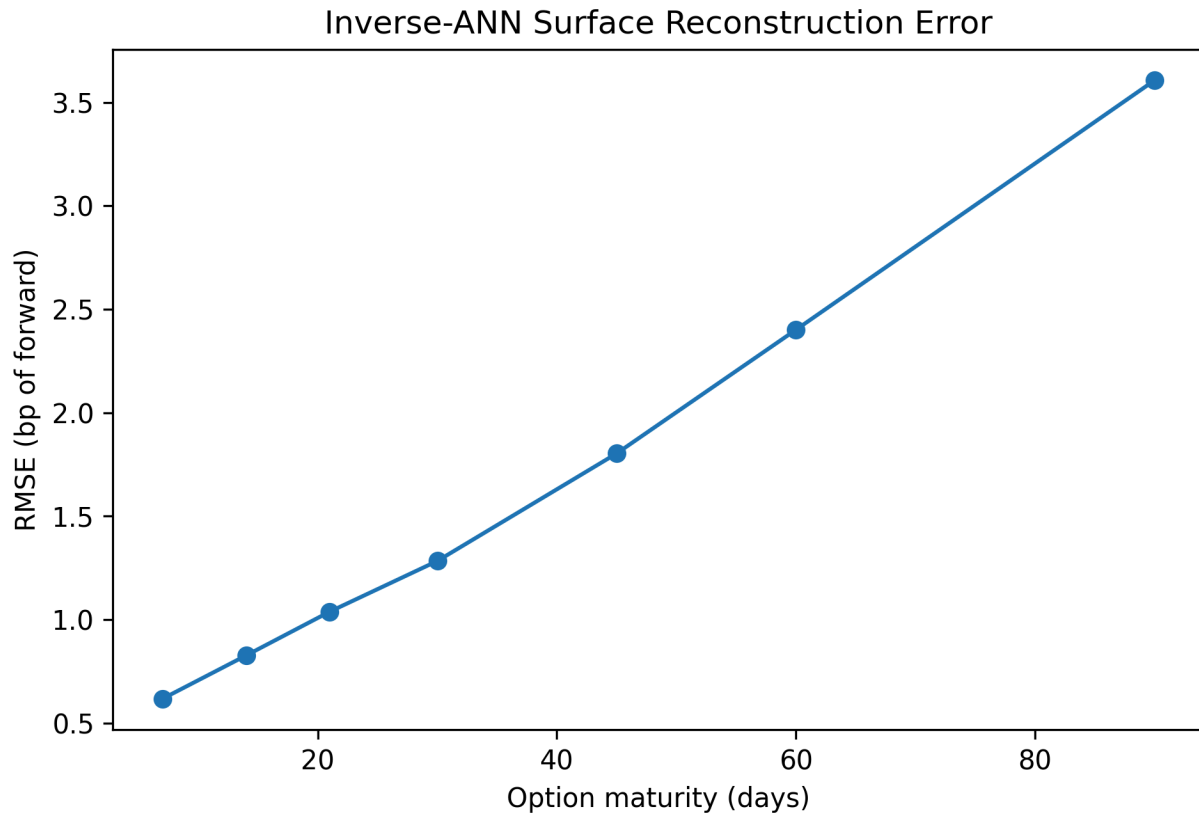


Figure A2. Root-mean-square error between original synthetic PDV option surfaces and surfaces reconstructed using inverse-neural-network parameter estimates. Error increases with maturity but remains at the basis-point scale across the 7-90 day grid.

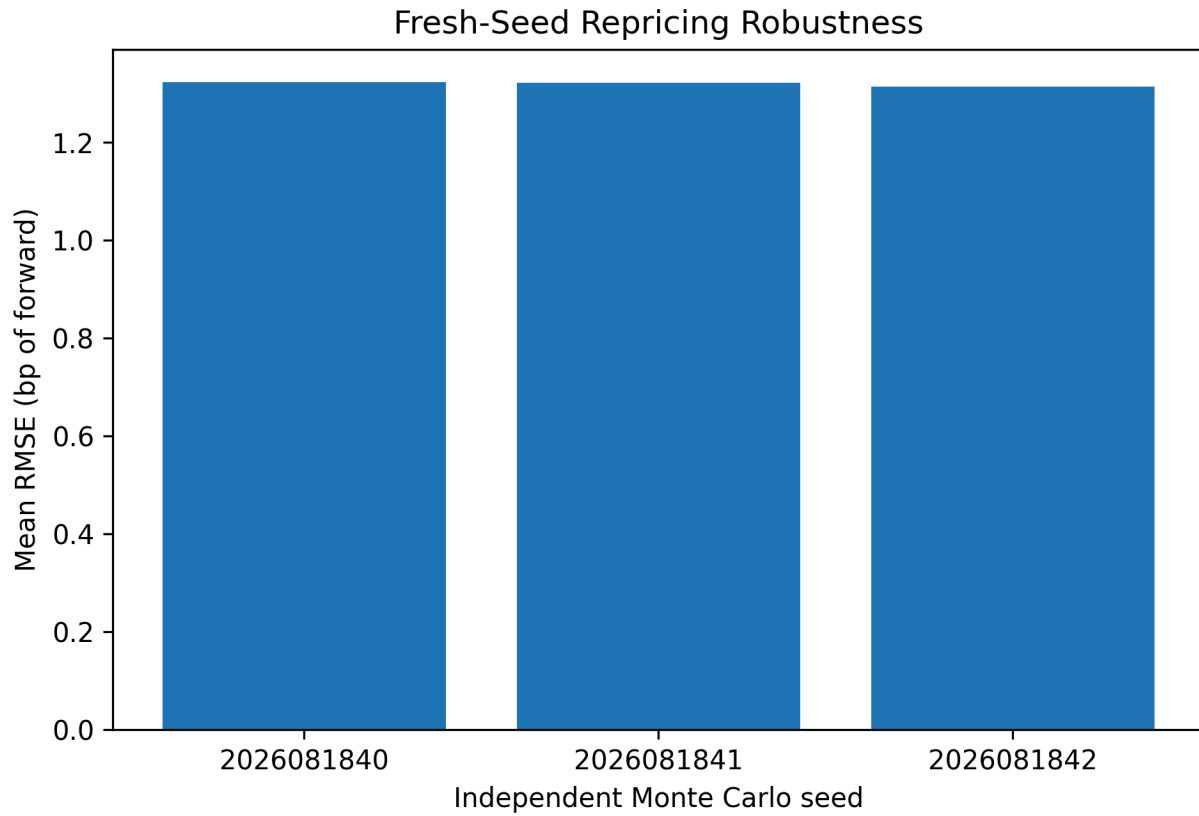


Figure A3. Mean surface-repricing RMSE for 64 balanced test scenarios under three previously unused Monte Carlo seeds. Similar errors across all three seeds indicate that reconstruction accuracy was stable across independent simulation realizations.

Appendix B: Parameter Recovery Diagnostics

Figures A4-A8. True versus inverse-neural-network predicted PDV parameters on the untouched 959-scenario in-domain test set. The diagonal line denotes perfect recovery. Test-set R^2 values were 0.9934, 0.9883, 0.9943, 0.9689, and 0.9933 for β_0 , β_1 , β_2 , θ_1 , and θ_2 , respectively.

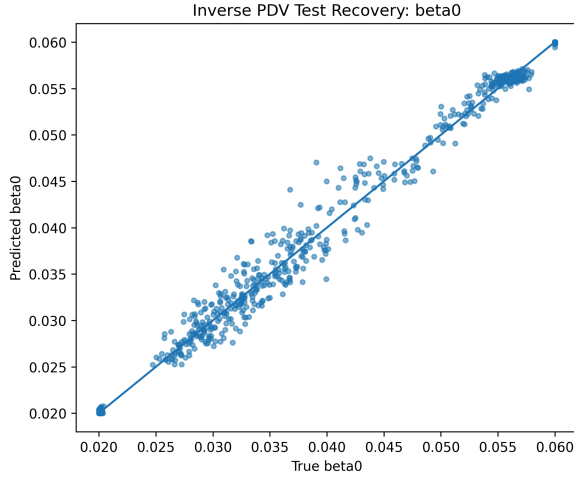


Figure A4. β_0 recovery.

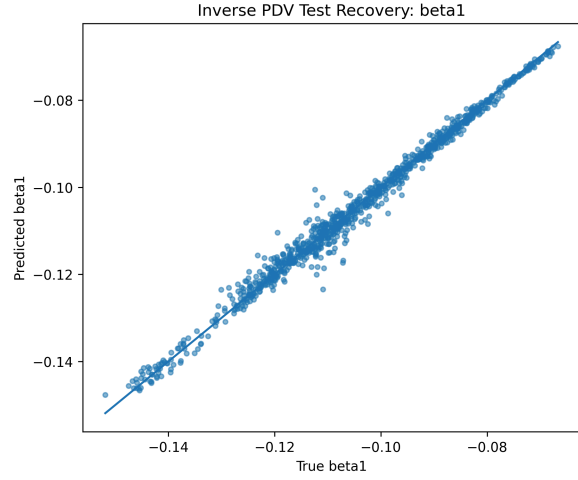


Figure A5. β_1 recovery.

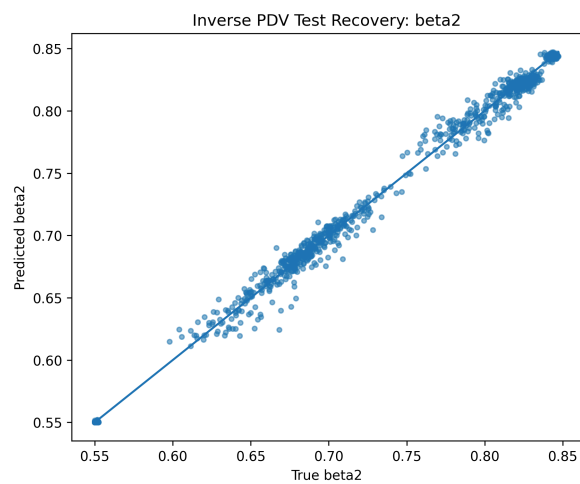


Figure A6. β_2 recovery.

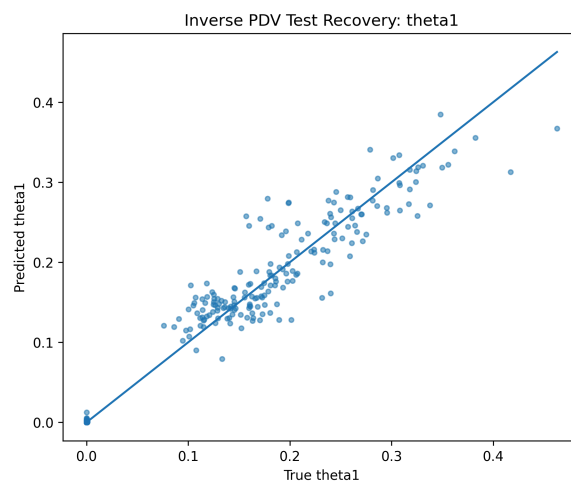


Figure A7. θ_1 recovery.

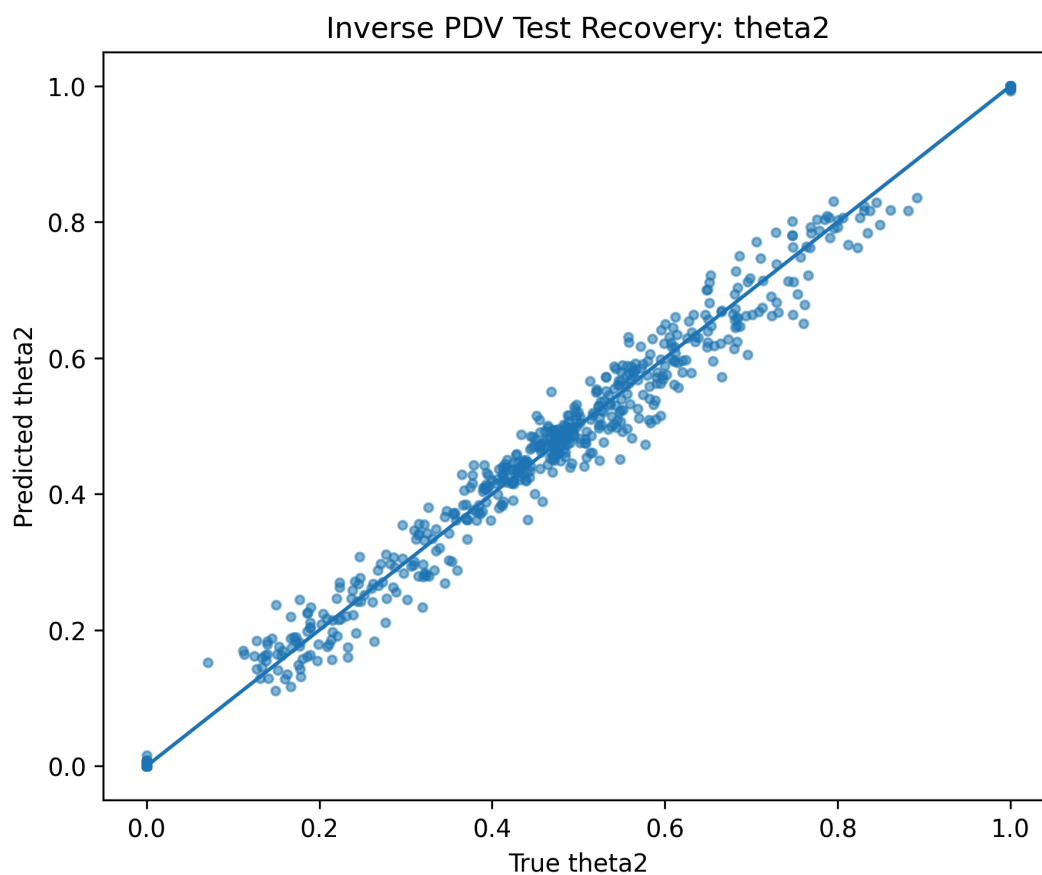


Figure A8. θ_2 recovery. True versus predicted values on the untouched in-domain test set; the diagonal line denotes perfect recovery.